

CS5182 Computer Graphics Revision



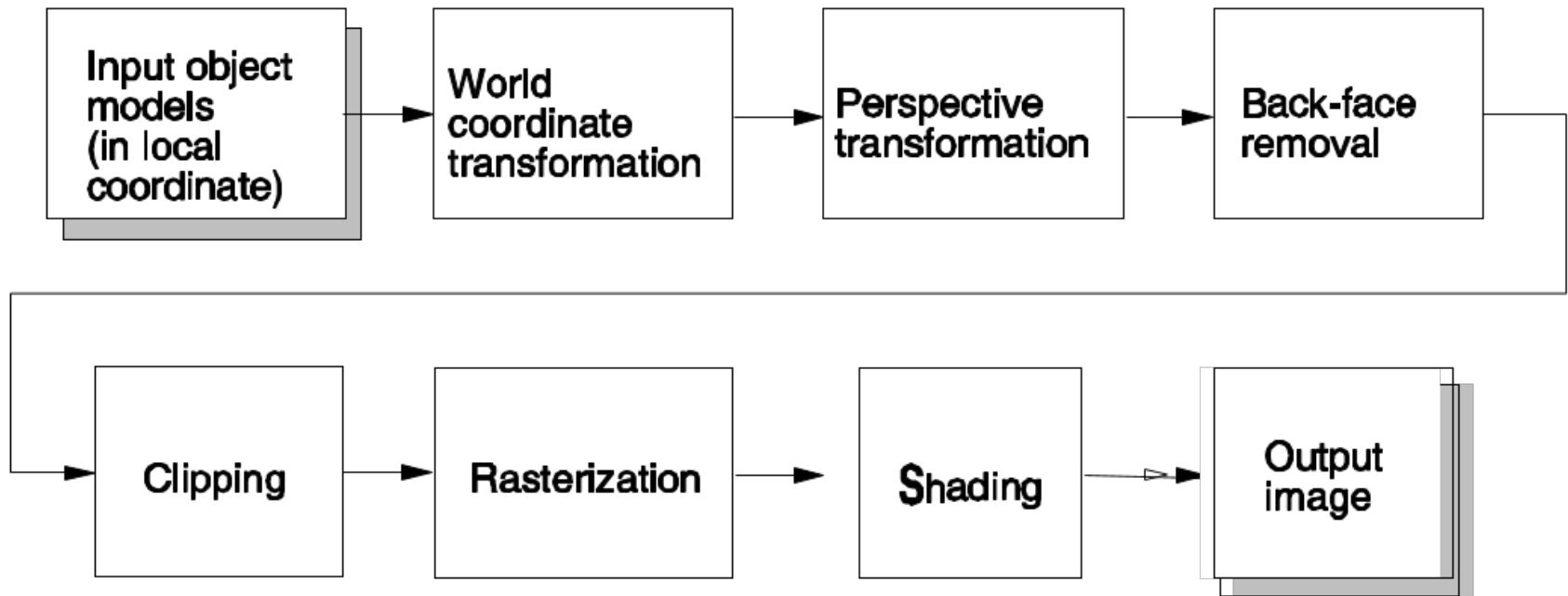
2025/26 Semester B

Department of Computer Science
City University of Hong Kong

Three Types of Rendering

- ❑ Rasterization-based rendering pipeline
- ❑ Ray tracing and radiosity
- ❑ Real-time Rendering
- ❑ Neural rendering (NerF) and 3D Gaussian Splatting (Tutorial) This part is NOT required in the final exam.

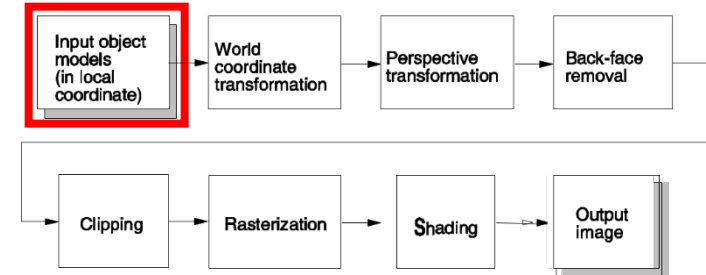
Rasterization based rendering pipeline



Rasterization-based rendering pipeline

□ Object modeling

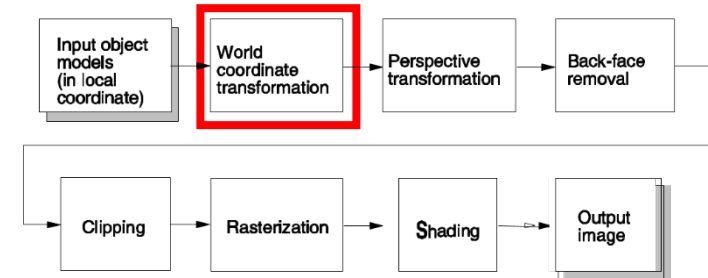
- 2D drawing--- a line, a circle
- Representations of 3D object
 - Point clouds
 - Polygonal meshes (orientable/non-orientable, Euler formula)
 - Subdivision surfaces
 - Implicit surfaces
 - Parametric surfaces
 - Voxel
 - Constructive solid geometry
 - Fractals



Rasterization-based rendering pipeline

□ Transformation

- Homogeneous coordinates
- 2D transformations
 - Translation
 - Scaling
 - uniform/non-uniform, simple/general case
 - Rotation
 - rotation orientation, simple/general case, three-step trick
 - Shearing
- 3D transformations
 - Translation, scaling, rotation around x, y or z-axis
- Inverse transformation
- The order of transformations



Rasterization-based rendering pipeline

□ Projection

- The elements of projection:

- COP, projector, projection plane

- Parallel projection

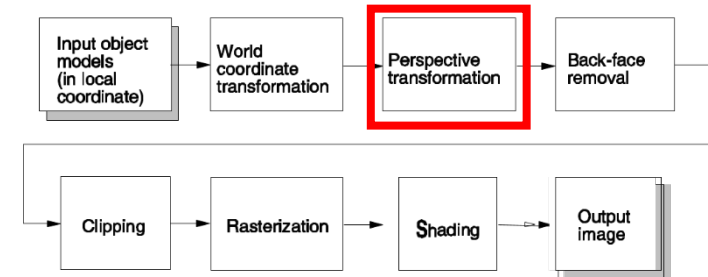
- Orthographic projection and the corresponding projection matrix
 - Oblique projection

- Perspective projection

- Vanish point
 - One/two/three-point perspective projection
 - Perspective projection matrix

□ Perspective transformation

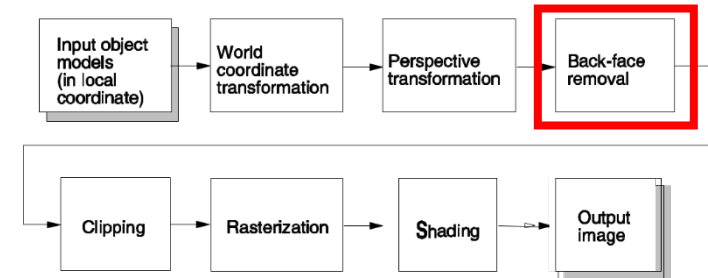
- After the transformation, the z coordinate keep unchanged
- Its advantages
- The transformation matrix



Rasterization-based rendering pipeline

□ Hidden/Back Surface Removal

- Object space algorithms
 - Back-face culling
- Image space algorithms
 - Depth-sorting (or Painter's) algorithm
 - Z/Depth-buffering algorithm



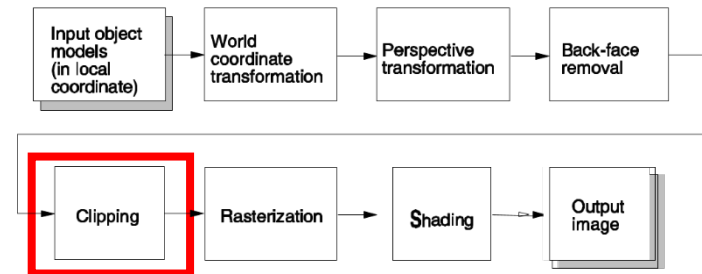
□ After the perspective transformation

- Using the z component of the normal vector of a typical face

Rasterization-based rendering pipeline

□ Clipping

- The Cohen-Sutherland line-clipping
- The Sutherland-Hodgman polygon-clipping



□ After the perspective transformation

- The 6 clipping planes become parallel to the three axes, making the clipping easier.

Rasterization-based rendering pipeline

□ Rasterization

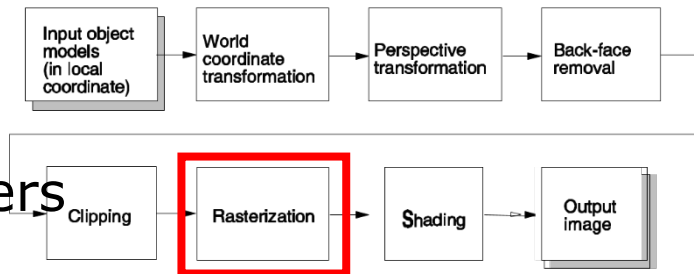
- The process of taking a primitive and figuring out which pixels it covers

□ Aliasing

- Spatial aliasing
- Temporal aliasing
 - Induced by the spatial aliasing
 - Undersampling in the time domain

□ Anti-Aliasing

- Supersampling
- Accumulation buffer
- Stochastic sampling
- Catmull's algorithm
- The A-buffer method



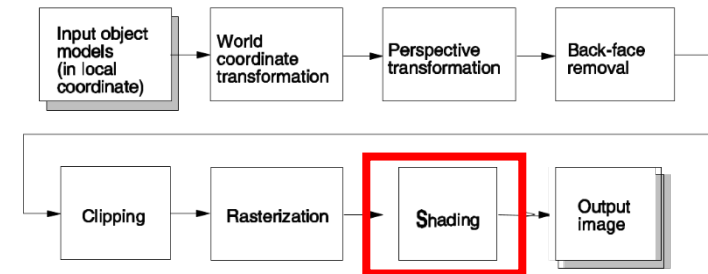
Rasterization-based rendering pipeline

□ Illumination model

- Light sources
 - Ambient/point/area/spotlight
- Phong illumination model
 - Ambient reflection
 - diffuse reflection
 - specular reflection

□ Shading method

- Flat shading
 - Gouraud shading
 - Phong shading



Ray Tracing and Radiosity

□ Ray Tracing

- The process
- How to calculate the color of a pixel
- The advantages and disadvantages
- Acceleration
 - Bounding volumes
 - Space subdivision

□ Radiosity

- The process
- The advantages and disadvantages

Real-Time Rendering

- Progressive rendering technique
 - Discrete LoD
 - Progressive mesh
 - Edge collapse
 - Vertex split
 - Selective refinement
- Ways to estimate the visual quality and rendering cost
- Shadows
 - Hard shadows
 - Shadow volume and shadow map
 - Soft shadows

GPU and Animation

□ GPU

- GPU vs. CPU
- Efficient computation: Data-level parallelism and task-level parallelism
- Efficient communication
- GPU architecture
 - Traditional hardware graphics pipeline
 - Advanced hardware graphics pipeline (Programmable vertex processor, rasterizer, programmable fragment processors (which processor performs which tasks?))

□ Computer Animation

- Key frame
- Procedure
- Motion capture

Point Cloud Processing

- The irregular structure of point cloud data makes it challenging to design deep neural networks
 - How to conduct convolution
 - How to find neighbors
 - How to aggregate features
 - Several popular feature extraction frameworks
 - Pointnet
 - Pointnet++
 - Dynamic graph cnn

In the final examination, there are **NO** questions for this section.

Final Examination

- There are **TEN** questions. **Closed-book**
- Question type
 - Understanding of basic algorithms and operations introduced in the lecture. (~24%)
 - 2 questions about transformation requiring simple calculations. (~33%)
 - Ray tracing, illumination model, shading, aliasing (~43%)
 - All questions are designed based on the content of the lecture notes and explanations in the lectures.